Karthik Viswanathan

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I am pursuing a PhD at the University of Amsterdam, focusing on **interpretability** of large language models through **geometric analysis** of their internal representations. My interdisciplinary background began with **competitive programming**, progressed through developing machine learning models as a **surveillance analyst** at Goldman Sachs, and was enriched during my master's in **theoretical physics**. My cross-domain exposure has allowed me to transition smoothly across disciplines, enabling meaningful contributions to both practical implementations and theoretical investigations in scientific research.

Education

University of Amsterdam — **Amsterdam, Netherlands**
PhD in Physics, Expected: December 31, 2025 — *2021–2025*
Thesis: From Language Models to Cosmic Structures: A Geometric Perspective
Advisor: Dr. Jan Pieter van der Schaar

University of Amsterdam — **Amsterdam, Netherlands**
MSc in Physics and Astronomy (Theoretical Physics), cum laude — *2019–2021*
Thesis: *Exploring the Spectral Theory/Topological Strings duality*
Advisor: Dr. Marcel Vonk
GPA: 8.7/10

Indian Institute of Technology Madras — **Chennai, India**
B.Tech in Engineering Physics — *2013–2017*
Thesis: *Real Space Renormalization and Applications to Machine Learning*
Advisor: Dr. Ashwin Joy
GPA: 8.29/10

Research Experience

Probing Geometry of Next Token Prediction Using Cumulant Expansion of Softmax Entropy
Karthik Viswanathan, Sang Eon Park
ICML 2025: Workshop on High-dimensional Learning Dynamics

Persistent Topological Features in Large Language Models
Yuri Gardinazzi*, **Karthik Viswanathan***, Giada Panerai, Alessio Ansuini, Alberto Cazzaniga, Matteo Biagetti
Proceedings of the 42nd International Conference on Machine Learning — **Equal contribution*

The Geometry of Tokens in Internal Representations of Large Language Models
Karthik Viswanathan, Yuri Gardinazzi, Giada Panerai, Alberto Cazzaniga, Matteo Biagetti
Under review

Cosmology with persistent homology: A Fisher forecast
Jacky Yip, Matteo Biagetti, Alex Cole, **Karthik Viswanathan**, Gary Shiu
Journal of Cosmology and Astroparticle Physics (JCAP), Vol. 2024, Issue 09

Workshop: Interpretability in LLMs using Geometric and Statistical methods
Karthik Viswanathan, Jan Pieter van der Schaar
Organized a two-day workshop with around 40 participants from Amsterdam and AREA Science Park (Trieste).

Professional Experience

Goldman Sachs **Bangalore, India**
Surveillance Analyst *2017–2019*
Developed ML models for anomaly detection in financial data
Manager: Prof. Howard Karloff

Goldman Sachs **Bangalore, India**
Summer Intern *2016*
Implemented fast Personalized PageRank in a MapReduce framework
Mentor: Dr. Koushik Balasubramanian

Awards & Distinctions

Sander Bais Prize for Academic Merit **Amsterdam, Netherlands**
Institute for Theoretical Physics Amsterdam *2020*
For exceptional academic performance in the master's program

Honorable Mention **Marrakech, Morocco**
ACM ICPC World Finals *2015*
Represented India in the international collegiate programming contest.

Participant **Bangalore, India**
International Olympiad in Informatics Training Camp *2013*
Selected after ranking in top 25 in the Indian National Olympiad in Informatics

Organization & Mentoring

AREA Science Park, Trieste
Master Student Supervisor *2024*
Student: Giada Panerai
Thesis: Zig-Zag Persistence in Neural Networks Representations

University of Amsterdam
Master Student Mentor *2023*
Student: Sibilla Bouche
Thesis: The persistence of non-Gaussian features: a Neural Ratio Estimation approach

University of Amsterdam
Master Student Supervisor *2023*
Student: Sliem el Ela
Thesis: From Primordials to Persistence: A Dual Exploration of Multiparameter Topology and Cosmic Origins

University of Amsterdam
Journal Club Organizer *2022*
Cosmology Journal Club

Technical Skills

Programming: PyTorch, TensorFlow, Hadoop, MapReduce, Mathematica

HPC Resources: LUMI consortium, Snellius supercomputer

Teaching

University of Amsterdam **MSc Physics Program**
Teaching Assistant *2020–2024*
Courses: Topological Data Analysis: a Physics Perspective, Advanced Cosmology, ML for Physics and Astronomy, QFT in Curved Spacetimes, General Relativity

Research Visits

2024: **AREA Science Park**, Trieste, Italy September 16 – October 4
2024: **AREA Science Park**, Trieste, Italy February 15 – April 27

References

Dr. Jan Pieter van der Schaar
University of Amsterdam
`j.p.vanderschaar@uva.nl`

Dr. Magnus Botnan
Vrije Universiteit, Amsterdam
`M.B.Botnan@vu.nl`

Dr. Matteo Biagetti
AREA Science Park, Trieste
`matteo.biagetti@areasciencepark.it`

Dr. Alberto Cazzaniga
AREA Science Park, Trieste
`alberto.cazzaniga@areasciencepark.it`